Reflections on Operators' and Maintenance Engineers' Experiences of Smart Factories

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ABSTRACT
In production environments, the number of distributed, networked, and automated systems has grown rapidly and is expected to continue to grow in the future. This affects humans' work fundamentally, in terms of their tasks and routines. Increasing automation and digitalization leads to a substantial change of human-machine interactions on the shop floor, raising the question about humans' role in highly automated environments. In this paper, we shed light on how work in increasingly automated and digitalized factories is experienced, drawing on interviews with operators and maintenance engineers from three different industrial contexts. By reflecting on actual and anticipated developments in smart production environments, we point out how workers will experience those contexts. We finally discuss resulting challenges and leverage points for smart factories, i.e., areas where HCI and CSCW can contribute to positively influence workers' experiences in times of increasing automation and digitalization.

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Smart Factory; Factory of the Future; Industry 4.0; Work; Experience.

INTRODUCTION
Modern production environments are characterized by increasing automation and digitalization. Systems in the factory are interconnected with other systems and working procedures are more and more automated. This change does not only concern production modes and routines, it also affects humans' everyday working routines in fundamental ways. The trend towards digitalized, networked, and automated systems is referred to as *smart factories, factories of the future, fourth industrial revolution,* or *industry 4.0*. Thereby, an academic as well as a public discourse started to take place, discussing economic, social, societal, and political consequences.

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What is less discussed is how humans' interactions with systems and technologies are affected by these changes. Systems are increasingly interconnected and may be difficult to separate. Systems' operations become more complex and are harder to understand; processes are automated and cannot be intervened easily. Accordingly, workers' tasks and routines may need to be adapted, completely changed, or may even become obsolete. However, what does such a change of work mean for humans' role in production environments? How is the relation of humans and systems configured? How do humans experience working in increasingly automated and digitalized production environments?

This work contributes a reflection on workers' experiences in highly automated and digitalized factories. In particular, we explored how increasing automation and digitalization is experienced and anticipated by workers on the shop floor, relying on interview data from 10 operators and 22 maintenance engineers from three different industrial contexts in Central Europe. Based on our results, we point out areas where research and design can play a considerable role in promoting positive experiences in smart factories. These areas address control, workload, qualification and knowledge, as well as involvement and responsibility. Looking beyond single interactions in the factory, we envision the factory context from an experiential, holistic perspective [52], pointing out challenges and potentials of the fourth industrial revolution from an experiential point of view.

Our paper is structured as follows. First, we present related work regarding smart factories to demarcate and motivate our work. Next, we describe our research context, as well as the applied method and analysis. Then, we describe characteristics of work building on interviews with operators and maintenance engineers. Finally, we reflect on the results, point out challenges and leverage points to enhance workers' experiences, and discuss our insights.

BACKGROUND
In this section, we provide a brief overview of the historical development of manufacturing, point out promises and visions coming along with increasing digitalization and automation (including potential consequences for humans), and indicate how Human Computer Interaction (HCI) and Computer Supported Cooperative Work (CSCW) are involved in these developments.

A Historical Perspective on Fabrication
When considering the evolution and development of fabrication from a historical perspective, four industrial revolutions

can be distinguished [43]. Beginning in the late 18th century, the first industrial revolution used water and steam power to mechanize production. Textile industry was the first to be affected by it. Tasks previously done laboriously by hand in hundreds of weavers' cottages were brought together in a single cotton mill and the factory was born. The second industrial revolution used electric power to create mass products. With moving assembly lines and the division of labor, mass products could be fabricated. With the third industrial revolution, electronics and information technology were used to automate production: digital technology found its way into factories.

Now a fourth industrial revolution is becoming apparent. This current revolution is characterized by a fusion of technologies, blurring the lines between the physical, digital, and biological spheres [43]. So-called cyber-physical systems (CPS) are introduced in traditional industries, meaning that many purely mechanical devices have evolved into computer-controlled and networked electromechanical systems [9]. The Internet allows communication between humans and machines in cyber-physical systems throughout large networks [8]. Whereas the first, second, and third industrial revolution are describing historical evolutions, the fourth industrial revolutions is rather a future vision or direction for factory environments, based on current technological developments and opportunities.

Smart Factories: Promises & Visions

Referring to the above mentioned fourth industrial revolution, the German government introduced the term *Industry 4.0* in 2011. This term summarizes "all technologies, trends and developments which serve to advance production plants" [47, p.1] and emphasizes the development of factories towards automation and digitalization. Similarly, the terms *smart factory* or *factory of the future* represent keywords to denote highly technological developments of factories [8]. In the following sections, we explain what increasing automation and digitalization mean from a technology-focused as well as from a human-focused perspective.

Technology-Focused Perspective

According to Broy and Schmidt [9], highly technological developments involve cyber-physical systems (i.e., control of mechanical activity through embedded computing) and advanced visualization techniques of data via virtual reality. Further, systems are aware of the environment and other objects through sensors and can interact with the environment through actuators. Thus, systems are dynamic and context-sensitive. Networking and connectivity are other important characteristics regarding smart factories. Systems are no longer located at a specific, unique place but distributed over several places, communicating with each other via networks. Smart factories come along with visions and promises of virtualization, decentralization, and network building [8].

Technological challenges associated with the new developments include individualized production (e.g., personalized mass products, modularization), collaborative networks (e.g., networks of legally independent organizations that share competencies in order to exploit a business opportunity), or end-to-end digital integration (e.g., advanced methods of communication and virtualization) [8]. Unquestionably, these technological developments already affect and will affect modern production environments and their socio-economic contexts. Thus, a range of disciplines is concerned with factories and their future, shaped by different foci and research interests: economics and sociology (e.g., [10, 19, 25]), computing, engineering and technology (e.g., [8, 9, 47]), or human factors and ergonomics (e.g., [12, 21, 22, 41]).

Human-Focused Perspective

With regard to human actors, different structural levels like shop floor, engineering, or management can be distinguished. Each of these structural levels is characterized by specific socio-technical constellations and system requirements. In the light of increasing automation and digitalization, it is uncertain if humans are still needed in smart factories. A common opinion is that humans will still be necessary; however, work and tasks will change [8]. This change could result in an *upgrading of qualifications* through the automation of simple and low-level tasks [55]. This can be considered as *skill-biased technical change*, meaning that the winners of increased automation are those who have already higher qualifications and resources [10]. In particular, there is an increasing need for coordination [8], requiring operators on the shop floor to be skilled in decision making (e.g., to solve problems in case of unforeseen events). The role as decision maker or coordinator comes along with an increase of work content for the single worker, making experience-based trouble shooting a main task of workers in future factories [25].

The increasing gap between complex tasks requiring high-qualified workers and simple tasks requiring low-qualified workers, with decreasing importance of qualifications on a mid-level, is considered as *polarization of qualifications* [25]. Jobs requiring such mid-level qualifications will be unnecessary or substituted by automation. In contrast to the upgrading hypothesis, simple activities will not disappear but new simple activities will evolve with increasing automation [10]. Overall, a polarized work environment will evolve, which could be considered as an environment which provides either *lousy or lovely jobs* [19].

Regarding humans in relation to machines, different philosophies exist. According to the *prosthesis design* philosophy, the machine expert is in complete control of the processes [38]. This means that the human is no longer needed to operate processes or carry out tasks. Considering the *human in the loop* philosophy, the human can actively intervene in processes. That means that the typical *safety valve* to manage problems is to keep a person "in the loop", requiring the person to apply his or her expertise in making the final decision on what actions to take [45]. Another philosophy considers *humans and machines complementing each other* [12, 25]. The effectiveness of such joint systems depends on the ability of both human and machine agents to coordinate and capitalize upon the unique abilities and information to which each agent has access [12].

The present paper takes a human-focused perspective, investigating and reflecting on workers' experiences in smart factories. Thereby, the above mentioned philosophies represent perceptions of human-machine relations which may be indica-

tive for workers' experiences of smart factories. Relations between humans' experiences in smart factories and the above stated philosophies should be explicitly considered and discussed for that reason.

Human-Machine Interaction in Factories

Especially the CSCW community emphasizes the importance of investigating new technologies in the field (for an overview see [3]). For example, Brahe and Schmidt [6] ask for investigating how workflow technologies "are developed, deployed, appropriated, amended, used, redeveloped in the wild". Or, as Harrison and Dourish [23] point out "The placeness [...] is an evolved set of behaviours rooted in our ability to creatively appropriate aspects of the world, to organise it, and to bend it to our needs", emphasizing the shared understandings of appropriate use, and the social interpretation of cues in the physical environment. Regarding recent developments like for example digital fabrication, Ludwig et al. [31] also point out the importance of investigating practices of appropriation in the field to make technologies more sociable. Considered from a historical perspective, the so-called Francophone Ergonomics tradition put workers in the centre of work, paying particular attention to field studies, especially focusing on the effects of increased mechanization and automatic control systems. Attention was paid to working conditions of highly automated control systems as well as cognitive and organizational requirements related to decision making in technically complex work settings (e.g., supervisory process control in time- and safety critical work) [3].

Research on CSCW often focuses on work practices and work organization in a variety of contexts. The integration of IT systems or workflow management systems in relation to existing work practices and work organization was for example investigated in medical or financial contexts (e.g., [6, 20, 39]), as well as in the context of trouble shooting at call centers [11]. Other studies in the medical context investigated work organisation related to roles and practices (e.g., [5, 42, 46]. Collaborative practices and the integration of technology was also studied in the context of aircraft [50]. Further topical areas of CSCW research are mobile knowledge work [15] or crowd working [24, 35]. Summing up these studies, they point out the importance of aspects like control, communication, situation awareness or involvement of workers as well as the possibility of active intervention. Further, they emphasize the need for organizational implementation of new ways of working, the configuration of work as a means to provide influence and responsibility for work, or the awareness and visibility regarding changing work practices when introducing a new technology at work.

Previous work in human-centered automation and man-machine studies (e.g., [1, 38, 44, 45]) discusses potential consequences for workers regarding future factories. One such consequence is for example that simple presentations of low-level data become insufficient as they produce high cognitive load due to rising complexity and autonomy, considered as "data availability does not equal informativeness" [12, p.5]. It is essential that the human maintains an understanding of the problem from the system's perspective (e.g., solution tactics) to let humans intervene appropriately in critical situations [12, 38]. According to Hirsch-Kreinsen [25], increasing complexity of systems is related to a limited controllability of systems, coming along with functional and economical challenges and unanticipated requirements due to lacking experiences of workers. This leads to a so-called *functional and informational distance* between workers and systems [25], which makes it difficult for workers to accumulate new knowledge, control systems, and take over responsibility of system processes. Digitalization and automation may lead to a loss of practical (tacit) knowledge, experience, and problem solving skills (which is especially relevant in case of unexpected errors). For example, monitoring tasks without direct contact to physical and material processes on the shop floor lead to an elimination of "the informal feedback associated with vibrations, sounds, and smells that many operaters relied upon" [28, p.419]. Further, detailed instructions regarding tasks and working procedures make detailed knowledge unnecessary and convey a loss of know-how [25]. Hancock [22] considers operators' overreliance on automation as *automation bias*, pointing out that employees and management have too much trust in the ability of digital systems to automate and support work procedures. Especially in case of unexpected (and mostly inevitable) incidents, highly automated processes lead to problematic situations which are difficult to solve and hard to cope with due to their routinous nature. Already in 1983, Bainbridge [1] considered this as *ironies of automation*.

Smith [45] describes this as *brittleness*, i.e., the occurrence of situations and incidents which cannot be handled by the system, requiring a person to be in the loop (i.e., to apply his or her expertise in making the final decision on what actions to take). Thus, the human's role is to amplify the machine's adaptability of dealing with unexpected conditions [38]. This was also shown by Meneweger et al. [33], pointing out the necessity of humans to adapt to the automated systems and compensate failures in processes or task procedures. Regarding difficult and complex problems or failure situations, the human is mostly left with his own resources without any tools. This can be related to the informational distance described above: attempts to take the user out of the problem-solving loop can inadvertently increase the user's burden by requiring to handle the difficult cases without the benefit of experience and the practice of solving the simpler ones [16, 51].

Building on existing research from the CSCW community as well as man-machine studies, we take a closer look on an experiential perspective of work in the following. Especially in highly automated and digitalized factory contexts, such a perspective is often missing, in particular regarding operators' and maintenance engineers' experiences when working with smart manufacturing systems.

User Experience in Smart Factories

Humans are expected to be needed in smart factories, though in changing roles and for changing tasks [4]. This is based on the assumption that traditional processes and structures may (partly) remain in companies due to technological limitations. Further, short-term orientation and organizational structures of companies impede long-term oriented innovation [25]. It is,

therefore, questionable if there will ever be a completely new production environment with no "ancient" structures. Rather, hybrid structures are more likely to characterize smart factories.

That means that different stages of development may exist next to each other: Fully automated/digitalized systems, partly automated/digitalized systems, as well as systems which are not automated/digitalized at all.

Since the human is expected to be an essential part of future factories (as mentioned earlier), it is important to discuss how the developments towards the fourth industrial revolution will change humans' working routines and practices in smart production environments. This change of working routines, in turn, strongly impacts how humans' experience related technology usage in everyday factory work. Within HCI, this experiential view is subsumed under the umbrella term *User Experience* (UX) [18] and is also relevant for human-machine interaction in the factory. Regarding daily factory work, the relevance of workers' UX (e.g., [34]), as well as contextual influences on it (e.g., [53]) - comprehensively considered as Contextual User Experience [53] - have already been shown. For industrial settings in particular, the need to understand industrial user experience [36] has been pointed out. Further, creating value for companies through UX research [48] and the importance of defining UX goals [26] have been emphasized.

Changing work routines and practices impose challenges for UX in smart factory work regarding meaningfulness [30], joy [22], and positive experiences when interacting with technology [26]. For example, increasing automation could create high cognitive load and lead to a feeling of excessive demand due to increased complexity of processes. Tasks requiring passive monitoring of processes could represent a lack of challenge and lead to a feeling of boredom. Collaborating with advanced systems like robots could change interaction and communication at the workplace [40] and consequently affects workers' experience.

Although there is some research on UX in the factory, research focusing on the understanding of operators' and maintenance engineers' experiences in smart factories is missing. Potential effects of automation and digitalization on the workers' experience, i.e., the experience of workers on smart shop floors, is considered as an increasingly important topic to be investigated by the HCI and CSCW community. In particular, we address the exploration of challenges and opportunities that research and design will be confronted with in smart factories.

EXPLORING FACTORIES IN THE WILD

For our reflections on smart factories, we are drawing on qualitative interview data gathered in three different industrial contexts: automotive, electronics, and logistics sectors. Our studies took place in three facilities, one developing systems and solutions for inbound logistics (e.g., warehouses), one automotive supplier, and one operating in the area of semiconductor fabrication, i.e., producing micro chips. The studies were conducted within two nationally funded projects where we investigated human-machine interactions at industrial facilities, which consider themselves as Industry 4.0 contexts (i.e., deploying digital, interconnected, and automated technologies). Within these projects, we focused on humans on the shop floor, e.g., on operators and maintenance engineers in factories, or stakeholders in close contact with them (e.g., shift leads, trainers).

Qualitative Interviews

The reason for focusing on employees involved in operating and maintaining activities is that their work (directly involved in the production activities) will be mostly affected by changes coming along with increasing digitalization and automation on shop floors [29]. We conducted two waves of interviews, i.e., two interview studies with the first interview study taking place in April 2014 and the second taking place in May 2016. In both studies, interviewees were affected by the introduction of a new system. This was used as a trigger for interviewees' narrations regarding their experiences and anticipations in the light of increasing digitalization and automation. The first interview study was conducted at the semiconductor production site, focusing on daily work of human operators in the chip production, aiming to explore how operators experience working together with a newly deployed automated (robotic) system on the shop floor. The second interview study focused on maintenance engineers from all three industrial facilities, investigating how they experience working together with an intelligent assistive system. The insights presented in this paper are based on a sample of 32 participants from all three industrial partners. All interviews lasted about one hour.

Within the first study, we conducted 10 narrative interviews. Narrative interviews are open-structured qualitative interviews that aim to evoke reports of personal experiences by the participants [17]. The people interviewed were affected by the deployment of an automated (robotic) system. The system deployed was a robotic arm, which is able to load and unload machines for chip production. For the production of micro chips, a waver must undergo several production steps where it is treated by different machines. The robotic arm puts the wavers into the machines and puts it out from the machines after the procedure has finished. Before the introduction of the robotic arm, these loading and unloading tasks were done manually by human operators. We interviewed seven operators, two shift leads, and one dispatcher, all working together with the automated (robotic) system. In the interviews, we focused on workers' experiences regarding the newly deployed system. We were interested in expectations regarding the new system, actual changes of working routines, as well as anticipations for future factory work considering increasing automation and digitalization. Participants' mean age was 40 years (SD = 12), ranging from 22 to 57 years. Eight workers were male and two female. Their average working experience in the factory was 16 years (SD = 11), ranging from 0.5 to 30 years.

Within the second study, we conducted 22 semi-structured interviews [17]. Eight people were interviewed at project partner A working in the logistics context; eight people at project partner B working in the automotive domain, and six people at project partner C working in the semiconductor factory. In all three industrial contexts, a prototype of an intelligent assistive system for supporting experienced and inexperienced main-

tenance engineers was deployed. The interactive prototype was implemented on a mobile device (tablet) which provided different functionalities for assisting maintenance engineers in their daily work such as step-by-step guidance for specific maintenance tasks or error diagnostics support. To evaluate the system, the participants had to perform standardized maintenance tasks in the respective context by using the prototype for assistance (e.g., step-by-step guidance was provided). Afterwards, we interviewed the 10 maintenance engineers being familiar with service activities (in the specific context), and the 12 employees not being familiar with the specific maintenance task (e.g., working at the IT department, persons responsible for training or documentation)i.e., stakeholders indirectly involved in maintenance. On the one hand, the interviews focused on the assistive system prototype and its potentials for work. On the other hand, we explored maintenance engineers' daily working routines as well as actual and expected changes of their work due to technological innovations and increasing automation at production sites. The mean age of the participants was 41 years (SD = 11), ranging from 18 to 62 years; 20 employees were male and two female. Their average working experience in the company was 14 years (SD = 9), ranging from 1.5 to 32 years.

Analysis

The qualitative interviews were audio recorded, transcribed verbatim, and analyzed following a thematic analysis approach [7]. This analysis approach is used to organize qualitative data sets by identifying different themes within the collected data. For our purpose, we aimed to structure and describe the participants' statements about their experiences with and anticipations regarding new systems by revealing the prevalent thematic issues facing an increasingly digitalized and automated industry context.

In a first step, we selected relevant data with regard to the question how employees anticipate and experience changes. Then, the collected data was structured by assigning initial categories. In a next step, the categories were summarized into more general themes that aimed to describe people's experiences and assessments regarding the deployment of automated systems or smart technologies. Overall, from an operator's point of view, four main themes, and from a maintenance engineer's point of view, five main themes were identified. Based on the identified themes in the qualitative interviews, the findings were then interpreted in relation to their relevance for research and design. The outcomes of these reflections are presented after the next section, which shows the main insights from the interviews.

FINDINGS

In this section, we provide insights on how work in smart factories is described from both operators' and maintenance engineers' perspectives. Quotations have been translated from german to english.

Operators' Experiences

Below, we present experiences of operators that came along with increasing automation and digitalization in the studied factories. Selected quotations of operators are denoted accordingly from O1 to O10.

Monotony and Complexity

Working with automated systems was often associated with monotony. Such monotony can either be seen negatively or positively. When considered negatively, working with automated systems was experienced as more repetitive and monotonous, which was associated with a decrease of mental effort. One participant (O10) put it like this: *"It has made work boring... because it simply withdraws our tasks"*. In contrast, some of the workers saw monotony in a positive way, as monotony was linked to a decrease of mental effort and workload. One participant stated it in the following way: *"It has reduced working speed. Many handles* [i.e., movements] *can be omitted* [...]. *That is positive from my point of view"*. Next to monotony, an increase of complexity was linked to the introduction of automated systems. Ever since the deployment of automated systems, some procedures were experienced as more complex than before. Workers had to adjust their behavior to the system's behavior, including the anticipation of or reaction to the system's behavior. This made some working routines (e.g., the deposit of specific tools) more complicated for workers.

Adaption and Control

Regarding the collaboration with automated systems, the question who adapts to whom or controls whom was arising. Often, operators pointed out the necessity to adapt to the automated systems. That means that in practice, the operators had to adjust their working routines to the systems' work. In order to anticipate the system's behavior and execute their work properly, the operators had to constantly focus on what the system was doing. Thus, the human could act as a back-up for the automated system in case of defects/errors, i.e., the human replaced the automated system in case of problems to ensure that the production continued. Such an adaption represented a constraint in the operator's behavior and workflow. In our case study, the deployment of automated systems led to physical constraints for the operators, i.e., a limitation of space. Workers had to take care where to position things as the new systems required physical space, which was not demanded before. For example, one participant (O6) remarked: *"And the space is also* [limited]... *at some point you don't know where to place things"*. Another issue linked to the need to adapt to the system is the feeling of control. When talking to operators, they often stated that they experience no possibility to contribute to or influence the system's behavior. This leads to a feeling of non-involvement and lack of control. One participant (O2) stated it in the following way: *"Regarding the robot* [...] *you don't have any influence"*. Similarly, another participant (O10) pointed out that *"you should not intervene in the robot's actions* [...]. *You stand alongside and look at it* [the system] *until it is finished"*.

Knowledge Acquisition and Trouble Shooting

Knowledge is a central topic when introducing new systems and technologies in a factory. Operators' knowledge on how to handle these systems is crucial in daily work. Acquiring knowledge and skills in handling new systems is especially

important for problem solving, as in case of problems direct interaction with the system is needed. When knowledge is not provided sufficiently in trainings, trial and error strategies are often applied on-site to learn more about the automated system. This was described as follows by one participant (O10): *I mean, I have to say, we found out a lot by ourselves, because at night no one is here and we are doing overnight shifts. This means we have to be able to find solutions by ourselves and, thus, we were able to fix some errors with the manual, the electronic one, thankfully. Also without the technicians"*.

As learning happens while doing, mainly targeting the most important functionalities of the system, informal knowledge is especially important. This means that knowledge about the automated system was gained by asking or observing others (mainly other operators as well as technicians). This was pointed out by the following statement (O1): *"There have been trainings, but most information had been shared from worker to worker"*. In this study, direct interaction with the automated system was closely linked to trouble shooting. This means that in case of errors and problems, the automated system had to be actively operated (i.e., the system's routines have to be interrupted). Thus, knowledge acquisition was especially important for problem solving and trouble shooting.

Emotions and Feelings

Regarding operators' emotions and feelings, there were different fears linked to working with automated systems. Often, fear of being replaced was mentioned. This means that operators were afraid that humans are replaced by machines and not needed any more in the factory. For example, one participant (O1) stated: *"Clearly, this is linked to fear, as it* [the robot] *should replace the human somehow"*. Automated systems were often considered as taking away tasks from the operators: The automated system carried out tasks that were previously conducted by operators. Another issue was fear due to uncertainty of future developments. It was often unclear for operators what the deployment of new systems meant for their work. One participant (O1) pointed out that the number of automated systems will steadily increase by stating the following: *"How is that going on? If it begins like this it can only get worse'*. However, the operators also reported ambiguous feelings towards to the robots, since they were aware that innovation was needed and emphasized the necessity to keep track with technological progress.

Maintenance Engineers' Experiences

In this section we describe experiences of maintenance engineers, and stakeholders involved in maintenance. Selected quotations of maintenance engineers and involved stakeholders are denoted accordingly from M1 to M22.

Connectedness and Complexity

Employees and workers associate smart factories with increasing connectedness of systems and higher complexity. One participant (M13) considered this increasing connectedness as one of the biggest challenges regarding smart factories: *"The overall connection of all systems coming from different subsystems* [...] *drawing conclusions somehow and provide them to others* [...]. *I think this is the great challenge now, which we hopefully can cope with passably"*. Increasing automation was linked to increasing complexity, or, as one participant (M7) mentioned, *"a high level of automation is coming along with many new requirements"*. There were also concerns regarding the comprehension of increasingly interconnected and complex systems. For example, one participant (M2) indicated that the *"comprehension of the system could get lost* [...] *however, this is essential for trouble shooting"*. Another issue contributing to complexity is the integration of existing - and often old - systems. For example, one participant (M13) stated that *"we need also solutions for old facilities. I think this is the challenge."* Related to the connectedness of systems is the possibility to access information constantly (e.g., in the cloud). The extensive amount of available information in turn increased complexity, making it more difficult to select and provide the right information.

System Control and Monitoring

The supervision of systems - in the sense of monitoring and controlling them - play an increasingly important role in smart factories. On one hand, this should prevent problems beforehand, considered as *"the attempt to recognize problems preventively"* (M10). On the other hand, supervision of systems means the remote access and control of systems. In particular, this could mean that one remotely logs into a system for *"accessing equipment via the laptop"* (M22). Condition monitoring is another term mentioned in this context. Therefore, the current state of a system was checked constantly, and, if there is a specific abrasion, this information was passed on or displayed. One participant (M17) mentioned that in the past years, the control over systems (i.e., remotely accessing and controlling them) had increased, which, in turn, allowed to better trace and document systems' operations and performance. This participant further reported about a feeling of being monitored by the system, e.g., through the increasing need to document tasks and activities. This was perceived as *"surveillance"* (M17).

Increased Mobility and Permanent Availability

Changes for employees were often induced by an increase of mobile technologies like smartphones, laptops, or tablets. In relation to this, higher convenience of work due to reduced ways was mentioned. Mobile technologies often reduced ways, as necessary information was available on the mobile device. For example, one participant (M9) pointed out that *"the deployment of tablets for maintenance has made maintenance engineers' work much more comfortable, respectively, they have to walk much less than before"*. Closely related to this was the possibility to work remotely. Often, workers could access systems remotely, thus also saving time and pathways. As one participant (M10) stated, the possibility to work mobile and remote also has drawbacks as physical activity and movement is reduced: *"Much of the things for which someone had to go somewhere is in the meanwhile possible without moving"*. Further, this provided the opportunity to work from everywhere - even from home or from the airplane. Such a permanent availability may provide advantages and drawbacks at the same time; one participant (M13) considered it as *"a blessing as well as a curse"*. Some of the participants complained about the permanent availability at work due to mobile

phones, as this would interrupt them during work. One participant (M22) stated for example: *"Regarding mobile phones, it's the permanent availability. That is necessary especially during the night shifts, I know, but if you are getting thirty calls on the phone you are frustrated, eventually"*.

Availability of Expertise and Documentation

In the interviews, knowledge, information, and documentation of knowledge were central topics regarding smart factories. One participant (M11) stated it like this: *"Having access to experts is getting increasingly important as technology is developing very quickly and new customers are constantly joining"*. Expertise in maintenance and trouble shooting related to technological developments and innovations was found to be an important aspect. Similarly, availability of information was another important aspect mentioned by the workers. Workers and employees were increasingly interconnected via networks (e.g., internet), which, in turn, makes it possible to access information easily. Such an availability of information provided advantages as well as disadvantages and, thus, had to be organized in a way that is not too complex but easy to locate.

Related to that was a growing need for (appropriate) documentation. With increasing automation and digitalization, there were a lot of new products which have to be documented. When documenting, products, technologies, processes and context had to be understood in order to properly explain and report. The organization of processes was related to documentation. As one participant (M11) stated, *"if the product changes, the documentation has to change, too"*, meaning that increasing digitalization did not only affect production and products, but also the way processes were reported. One participant (M12) pointed out a trend towards mobile documentation which could be realized via information units. Information units could be considered as chunks of information that can be put together modularly. In order to be successfully applied, such a new approach (i.e., information units) would require new structures and strategies which are modular and flexible.

Emotions and Feelings

Emotions and feelings were also mentioned regarding maintenance work in smart factories. The fear of increasing unemployment was often pointed out. Further, employees and workers reported about the loss of personal contact. Feeling monitored by systems was also an impression. One participant (M17), for example, stated *"this did not take place in the past, this kind of surveillance"*. Reliefs were also coming along with increasing automation. For example, one participant (M3) described that *"everything is going on faster, and everybody wants faster responses. The advantage is that you can send photos or contact the technician on-site. That is a relief"*. However, at the same time, stress had increased, e.g., due to constant availability. Stress associated with constant availability was often linked to smart phones which were used as a work tool. As one participant (M21) stated, *"the* [smart]*phone is a stress factor which we did not have in former times"*. Overall, permanent availability seemed to induce stress in workers and employees, whereas remote access to systems may represent a relief (i.e., reducing the need to be on-site).

Summary

Based on the interviews, we identified the following characteristics regarding *operators' experiences and anticipations considering smart factories*: (1) monotony and complexity; (2) adaption and control; (3) knowledge acquisition and trouble shooting; (4) emotions and feelings. From *maintenance engineers'* point of view, the following themes denoted their *experiences and anticipations related to smart factories*: (1) connectedness and complexity; (2) system control and monitoring; (3) increased mobility and permanent availability; (4) availability of expertise and documentation; (5) emotions and feelings. In the next section, we reflect on the above listed themes and point out challenges and leverage points for research and design in smart factories.

DISCUSSION & REFLECTION

The introduction of new technologies in work environments in the sense of increased automation, digitalization and connectedness of systems is related to a spectrum of experiences workers have with these technologies. In the previous chapter, we showed how operators and maintenance engineers of three production facilities experienced an increase in automation. In this chapter, we reflect on our results in regards to workers' experiences in smart factories and resulting challenges and leverage points related to technological developments as well as humans' experiences, values and meaningfulness of work. The topics described in this section represent central categories which we will summarize and discuss in the following with regard to challenges and leverage points for CSCW and HCI.

Control

In the interviews with the operators as well as in the interviews with the maintenance engineers, control was discussed throughout. From a maintenance engineers' point of view, control seemed to be mostly related to the control of systems, in the sense that the system state is monitored or that systems are increasingly controlled remotely. In contrast, for operators it was questionable who is in control of whom, as operators often had to adapt to the automated system or were not allowed or able to control the system. In the production facilities we researched, the maintenance engineers actively controlled the systems, whereas operators rather seemed to be passive with regard to autonomous systems in the factory. They hardly had any influence or control but rather had to adapt their routines to the system's actions. Thus, control can range from the extreme of having total control over the system to the extreme of being totally obedient to the system.

In line with related work, it has been reported that humans strive for a feeling of control. According to Rosso et al. [37], the feeling of personal control represents one source for making work meaningful. Work with little space for action and decision-making negatively effects a person's well-being and intellectual capacity [49]. In a broader sense, this can be considered as self-efficacy, i.e., individuals' beliefs that they have the power and ability to produce an intended effect [2]. Striving for control in relation to the introduction of new technologies was for example shown by Brahe and Schmidt [6], who studied the introduction of a workflow management system in a large financial institution. Regarding human-machine

interaction in the factory, Kaasinen et al. [26] mentioned a sense of control over the system as relevant to transform negative feelings like anxiety or uncertainty into positive experiences. This may be an especially important aspect when considering operators who had to adapt their working routines and practices to the automated system. As this passivity may convey feelings that one's work is not important or can be replaced by automated systems, fears and uncertainties were often mentioned in relation to increasing automation and digitalization. Increasing complexity (e.g., through networked, distributed systems) makes it more and more difficult to understand processes and limits the controllability of systems [25].

From a technological perspective, which is shaped by the possibilities and requirements of machines, the need to control systems will decrease with increasing automation and digitalization. This in turn may limit humans' feeling of being in control over the system. From an experience-centered perspective, a feeling of control is crucial for the workers' self-esteem and meaningfulness of work.

Challenges and Leverage Points

A pressing question is *how to give workers (a feeling of) control whilst not counteracting technological developments regarding increasing automation and digitalization.* The following examples provide suggestions of how to convey a feeling of control in workers.

- *Possibility for human interventions:* Enabling workers to overrule the system in specific cases and situations gives them a feeling of control towards the system. For example, in case of errors, also operators (not only maintenance engineers and technicians) should have the possibility (e.g., knowledge and permissions) to take actions.

- *Transparency of system activity:* Knowing what a system is doing (instead of experiencing it as a black box) is essential to control systems [38]. Information about the system's state and activities provides a feeling of control in workers. For example, displays that allow to track system activities support transparency and control.

Workload

Our interviews showed that aspects like monotony, complexity, or permanent availability might come along with increasing automation and digitalization. In particular, maintenance engineers reported about increasing complexity, coming along with a range of new requirements and challenges. They experienced processes and systems as increasingly complex. Related to the introduction of new technologies, some maintenance engineers felt stressed and overstrained due to permanent availability (e.g., permanent calls on the phone). With regard to operators, increasing automation of processes led to boredom as tasks got monotonous and repetitive, or workers had long waiting times between tasks. For both, operators and maintenance engineers, knowledge about systems and processes declines as a result of increasing automation. This is in line with existing literature, claiming that due to the omission of the first order feedback loop (e.g., drawing, milling, grinding, drilling, handling, etc.), the worker is able to attend to a wider segment of the total transformation process and the required domain knowledge becomes more extensive and more complex [41]. Machines will increasingly take over tasks which were conducted by humans beforehand, changing workers' processes and routines. Related to increasing automation, workload can vary from low to high, meaning that workers' experiences may range from boredom to excessive demand. According to Goos [19], increasing automation and digitalization creates a divide between jobs which are not challenging at all and jobs which are very challenging, pointing out the spectrum of how workload may be experienced.

From an experience-centered perspective, adequate workload and challenge are important for positively experiencing work. Ideally, humans are stimulated by work, i.e., work allows them to optimally apply their abilities in a way that neither overstrains nor underchallenges them.

Challenges and Leverage Points

A challenge is *how to contribute towards an individualized, optimal level of workload for workers*. Below, we provide exemplary suggestions how boredom as well as excessive demand can be reduced towards optimal, well-balanced workload, creating a flow experience [13].

- *Assistive systems to reduce overstrain:* Providing technological support in form of assistive systems can help to reduce cognitive load of human workers. For example, assistive systems can offer step-by-step instructions for specific tasks, support humans in trouble shooting, or store information which can be dynamically requested.

- *Challenge and dispersion against boredom:* Bringing new input for workers can counteract boredom induced by monotonous or repetitive tasks. Such a new input could be gamification [14] of work, i.e., playful or challenging approaches for workers in relation to their (and other workers') work activities.

- *Mental and physical training to compensate inactive times:* We suggest to use non-active times for facilitating mental and physical activities, having a positive effect on work (as workers are better informed and/or healthier). For example, online courses and trainings may contribute to further education of workers, or physical trainings might compensate long standing or sitting.

Qualification and Knowledge

In our interviews, knowledge and expertise were considered as important by both operators and maintenance engineers, especially with regard to errors and unexpected incidents. Both, operators and maintenance engineers pointed out that the need of handling errors will rise with increasing automation and digitalization. Further, it will get more difficult to handle errors due to increasing connectedness of systems and increasing complexity. For operators, knowledge on how to handle new systems was crucial especially when errors occur; however, they reported that trainings were often insufficient so that they had to acquire knowledge informally or by trial and error. For maintenance engineers, access to knowledge was also an important topic; they mainly accessed expertise remotely

or via documentation. The statements of both operators and maintenance engineers showed the importance of knowledge, pointing out the important and differentiating role of education and constant access to knowledge in smart factories.

In line with related work, failures of machines are a crucial element of industrial work [45]. Mostly, failures represent unexpected situations which have to be handled by the human, requiring the so-called human in the loop [25]. Being prepared to handle unexpected situations is closely related to vigilance, i.e,. the need for sustained attention over a longer period of time, e.g., required for air traffic control [32]. When referring to humans in smart factories, most of the time experienced, educated and highly qualified workers are meant [47]. This is related to increasing complexity and the fact that problem solvers and trouble shooters may become increasingly important. In other words, "The more we depend on technology and push it to its limits, the more we need highly-skilled, well-trained, well-practised people to make systems resilient, acting as the last line of defence against the failures that will inevitably occur." [4, p. 65]. In accordance with the polarization hypothesis [10], simple jobs will also remain, as new simple activities will evolve with increasing automation. Thus, the gap between highly qualified and low qualified activities will increase.

From an experience-centered perspective, knowledge on how to solve errors is crucial [11]. As shown in our interviews, workers strive for diversification and self-determination. At best, work should be more than troubleshooting or passively reacting to pitfalls and errors. When not having the necessary knowledge to counteract errors and troubles, workers are likely to feel helpless or frustrated. Ideally, humans have access to knowledge (e.g., on systems, processes, products) whenever necessary. Knowledge acquisition can be either facilitated on-site by sufficient training or by access to necessary information.

Challenges and Leverage Points

A challenging topic is *how to provide workers with necessary knowledge and information tailored to the individuals' expertise*. We indicate exemplary suggestions for design in order to offer workers the possibility to gather the knowledge or training they need or would like to have.

- *Flexible learning approaches:* Offering the possibility of further education for workers in a flexible manner promotes knowledge acquisition [27]. Such trainings should be available in different modes so that a worker can choose according to what type of training is most adequate. For example, eLearning could be used as a means to train workers, with the mode of on-site (directly on the system) as well as online training.

- *Dynamic access to knowledge:* Providing the possibility to access knowledge whenever needed supports workers in handling situations which require specific expertise and knowledge, expanding existing qualifications. For example, assistive systems could dynamically provide knowledge dependent on the specific situation as well as on the qualification of the worker.

Involvement and Responsibility

Involvement and responsibility seem to be an important issue of work in increasingly automated and digitalized contexts for both operators and maintenance engineers. For operators, feeling involved in decision processes and developments was crucial regarding the deployment of new systems; not involving them led to uncertainties and fears regarding future developments. For maintenance engineers, responsibility was often associated with stress, which was even increased due to permanent availability related to new technologies such as mobile phones.

Consequently, responsibility and involvement might be crucial for workers' experience, being in the tension of high responsibility and non-involvement. Involvement and responsibility in the sense of decision making and space for action is crucial for one's well-being and intellectual capacity [49]. Being involved and having influence does not only convey a positive experience, it further promotes an understanding of the complexity of the involved systems and increases acceptance of new systems [6, 54]. According to Grote et al. [21], responsibility should come along with the possibility to change something, otherwise it will be frustrating. With increasing complexity, responsibilities are often no longer directly assignable. Christoffersen and Woods [12] consider this as directability, asking who owns the problem.

From an experience-centered perspective, providing the feeling of responsibility can reduce fears and uncertainties. Too much responsibility, however, can be linked to overstrain and stress. Thus, involvement of humans should be given but has to be balanced.

Challenges and Leverage Points

An important question is *how to achieve that workers feel responsible while not being overstrained*. Below, we point out exemplary suggestions for design in relation to workers' experiences.

- *Participation:* Actively involving workers in decision processes and deployment of new technologies counteracts uncertainties, fears and rumors. Workers should not be passively confronted with technological innovations but actively involved in decision processes [33]. For example, design decisions can be discussed and iterated in workshops with workers who will be affected by the new system.

- *Transparency of processes:* Whenever changes in everyday work occur (e.g., change of existing structures, processes, or work flows; introduction of new systems; hiring of new workers), workers should be able to prepare themselves beforehand instead of being immediately confronted with the new situation. This preparation does not only include information about the time frame, potential changes or reasons for the change; it also includes necessary knowledge acquisition (i.e., training) to be optimally prepared for new challenges coming along with innovation.

Limitations & Contribution

We are aware that our insights are bound to the three industrial contexts under consideration. The proposed challenges are

therefore a first collection of topics related to human-machine interactions in smart factories. With this, we open up a special application context to a broader audience and discuss opportunities for experience-centered design in increasingly automated and digitalized contexts. Beyond this, the presented challenges and potentials may be also relevant for humans' experiences with automated systems in other contexts. For example, with regard to the application area of (semi-)autonomous driving, designers are also confronted with the challenge to design for optimal, well-balanced workload of drivers.

Envisioning the future of production needs to consider that technological innovation is intertwined with existing, even outdated, technology. Even in highly automatized and digitalized shop floors, humans are expected to be needed, though in changing roles and for changing tasks [4]. Next to high-qualified workers who plan, introduce, supervise, and maintain the systems, practically experienced operators are needed to compensate automation deficiencies which are very likely to occur in the future. Autonomy and self-organization of systems will probably be partly applicable in the factory [10], so that proven and traditional forms of work will likely also remain [25]. Even with increasing automation and digitalization, industrial work in the factory will probably never be fully automated and digitalized; rather, we will find mixed forms of automation and digitalization that design needs to face, raising the question of how to optimally unite human and system competences. Therefore, smart factories have to be considered in constant transition between the first and the fourth industrial revolution, i.e., between mechanical and cyber-physical production.

Our findings are mostly in line with related work (e.g., [1, 4]), showing for example that monitoring and error-handling tasks increase with increasing automation. We extend existing research by looking at smart factories from a human-centric perspective, taking into account how humans interact with technologies in smart factories. Therefore, we explicitly consider experiences of workers with (new) systems in the factory (e.g., boredom, excessive demand, or loss of control) and discuss them in relation to challenges and leverage points for enhancing work experience. This connection between future tasks on the shop floor, experiences of workers, and resulting challenges has not been explicitly discussed in related work on factories (of the future) so far. Such a linkage provides insights and promotes further research regarding humans' role, experiences and the relation of humans and systems in smart factories and other highly automated and digitalized contexts.

CONCLUSION

In this paper, we critically discussed the role and associated experiences of workers and presented design challenges for increasingly automated shop floors. Our reflections are based on interviews with 10 operators and 22 maintenance engineers from three different industrial contexts. In relation to the introduction of new technological systems, we had the opportunity to interview them about their work experience and anticipations of the future. We discussed challenges arising in smart factories, i.e., areas where design can contribute to positively influence workers' experiences in times of increasing automation and digitalization.

Overall, we understand increasing automation and digitalization in factories as having both advantages and drawbacks. Changing work routines and practices impose challenges for the design of systems, interactions, and work environments. We see the role of design to cope with the changing work routines and practices in a way that workers' experience is taken into account and shaped towards positive experiences [26]. An exemplary potential for experience-centered design in relation to the above mentioned challenges is to design for perceived control. This could be for example achieved by enabling human interventions or providing transparency of activities and processes. Designing for balanced workload is another experiential goal that could be addressed, e.g., by facilitating technological support, offering challenging activities, or supporting mental and physical training. Further, designing for information provision could positively influence workers' experiences, e.g., by offering further education or flexible learning, or by providing dynamic access to knowledge. Assigning responsibility to workers may be another way how design could positively shape experiences, e.g., by allowing active involvement of users or providing contextual information.

Assuming that we can positively shape workers' experiences, we stress the importance of a holistic perspective on smart factories. We have to be aware that the fourth industrial revolution comes along with economical, political, educational, and social changes. These changes provide challenges and opportunities that have to be considered when discussing smart factories in general and the role of design in particular. For example, jobs and working hours may become more flexible and dynamic. Thus, work performance may be defined in other ways, as the measurement of working hours may not be adequate any more. With more flexible jobs, individuality of workers can be better considered so that humans can work in a way that best fits their strengths. Jobs in the factory may probably change, making existing differentiations between roles of operators, maintenance engineers, or technicians outdated and requiring new job descriptions. Further, education may also be defined differently, with work as an inclusive space for education and training.

The above mentioned issues point out the challenges and potentials of the fourth industrial revolution from a human-centered view. We hope that researchers and practitioners step into the complex context of smart factories and engage in the design of workplaces of the future, affecting not only work but human life in all its facets.

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